

Institute of Actuaries of India

Subject CT5 – General Insurance, Life and Health Contingencies

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1.

$$a) {}_{10}p_{45} = l_{55}/l_{45} = 9557.8179/9801.3123 = 0.975156$$

$$b) {}_{10|10}q_{40} = (l_{50} - l_{60})/l_{40} = (9712.0728 - 9287.2164)/9856.2863 = 0.043105$$

$$c) \ddot{a}_{55:10} = \ddot{a}_{55} - v^{10} {}_{10}p_{55} \ddot{a}_{65} = 13.057 - 0.515363 * 10.569 = 7.610$$

[3 Marks]**Solution 2.**

$$\begin{aligned} EPV &= (50000 - 2500) * 12 * A_{[50]:5}^1 + 2500 * 12 * (IA)_{[50]:5}^1 \\ &= 570000 * (A_{[50]} - v^5 {}_5p_{[50]} A_{55}) + 30000 ((IA)_{[50]} - v^5 {}_5p_{[50]} (5 * A_{55} + (IA)_{55})) \\ &= 570000 * (.20463 - .747258 * 9557.8179/9706.0977 * 0.26092) \\ &\quad + 30000 (4.84789 - .747258 * 9557.8179/9706.0977 * (5 * 0.26092 + 5.22868)) \\ &= 570000 * 0.012634 + 30000 * 0.040427 \\ &= 7201.4164 + 1212.8142 \\ &= 8414.23 \end{aligned}$$

[6 Marks]**Solution 3.**

- Terminal bonuses are allocated when a policy matures or becomes a claim as a result of the death of the life assured.
- Terminal bonuses are usually allocated as a percentage of the basic sum assured and the bonuses allocated prior to a claim.
- The terminal bonus percentage rate will vary with the term of the policy at the date of payment.
- As the policy is being terminated, the terminal bonus rate is usually chosen so as to distribute all the surplus available to the policy based on asset share.
- Distributing available surplus as a terminal bonus delays the distribution of surplus and may allow the insurer to choose investments that are more volatile in the short term but are expected to be more profitable in the long term.

[5 Marks]

Solution 4.**a) Death Strain at Risk**

If a whole life policy is issued t years ago to a life then aged x with sum assured of S payable at the end of the year of death.

Let ${}_tV$ be the reserve at time t . Then we define the death strain in the policy year t to $t + 1$ to be the random variable, DS , say,

$$DS = \begin{cases} 0 & ; \text{if the life survives to } t + 1 \\ (S - {}_{t+1}V) & ; \text{if the life dies during } t \rightarrow t + 1 \end{cases}$$

The maximum death strain, $(S - {}_{t+1}V)$ is called the death strain at risk or DSAR.

b) Expected death strain

The expected amount of the death strain for a single policy is called the expected death strain (EDS) for that policy. This is the amount that the life insurance company expects to pay extra to the year-end reserve for the policy. The probability of claiming in the policy year t to $t + 1$ is q_{x+t} , so that:

$$EDS = q_{x+t} (S - {}_{t+1}V)$$

c) Actual death strain

The actual death strain for a single policy is simply the observed value at $t + 1$ of the death strain random variable, that is:

$$ADS = \begin{cases} 0 & ; \text{if the life survives to } t + 1 \\ (S - {}_{t+1}V) & ; \text{if the life dies during } t \rightarrow t + 1 \end{cases}$$

d) Mortality Profit

Mortality Profit = Expected Death Strain – Actual Death Strain

[7 Marks]

Solution 5.

$$\ddot{a}_{[50]:25} = (1 - A_{[50]:25}) / d$$

$$d = 0.06/1.06 = 0.056604$$

$$\begin{aligned} A_{[50]:25} &= A_{[50]} + v^{25} {}_{25}p_{[50]} (1 - A_{75}) \\ &= 0.20463 + 1.06^{-25} * 6879.1673/9706.0977 * (1 - 0.56535) \\ &= 0.27641 \end{aligned}$$

$$\ddot{a}_{[50]:25} = (1 - 0.27641) / 0.056604 = 12.78343$$

$$120,000 \ddot{a}_{[50]:25} = 1,534,011.6$$

[4 Marks]

Solution 6.

$$EPV = 10,000 * ({}_{1/12}p_{60} * 1/1.09^{1/12} + {}_{2/12}p_{60} * 1.002328/1.09^{2/12} + {}_{3/12}p_{60} * 1.002328^2/1.09^{3/12} + \dots)$$

$$1.002328^{12} = 1.028296 \text{ and } 1.028296/1.09 = .94339 = 1/1.06$$

$$EPV = 10,000/1.002328 * ({}_{1/12}p_{60} * 1/1.06^{1/12} + {}_{2/12}p_{60} * 1/1.06^{2/12} + \dots)$$

$$= 1,20,000/1.002328 * a_{60}^{(12)}$$

$$= 1,19,721.29 (\ddot{a}_{60} - 1 + 11/24) [@6\%]$$

$$\ddot{a}_{60} = 11.891$$

$$\text{Thus, } EPV = 1,19,721.29 * 11.3493 = 13,58,756.83$$

[4 Marks]

Solution 7.

All three indices are heavily influenced by the mortality rates at the older ages, and therefore may not be helpful if it is the rates at younger ages that are of interest.

i) Comparison of national mortality at different times

Crude death rate: The crude death rate is easy to calculate but does not take into account the age/sex structure of the population. It could therefore be misleading if the structure has changed significantly between the two times.

Standardised mortality rate: The standardised mortality rate takes account of the population structure. It should therefore be a good indicator, provided the same standard population is used in both calculations. However, it is more complicated to calculate since it requires a breakdown of the occupational rates for each age/sex group.

Standardised mortality ratio: This also takes into account the population structure and should be a good indicator. This requires data for the occupational population numbers in each age/sex group. [3]

ii) Comparison of mortality for different occupational groups

Crude death rate: As the crude death rate does not take into account the age/sex structure of the population and different occupations may have very different population structures, it would not be a reliable indicator for comparing the effect of different occupations on mortality.

Standardised mortality rate: As the standardised mortality rate takes account of the population structure, it should provide a good indicator, provided the same standard population is used in both calculations. Again, it requires age/sex-specific rates.

Standardised mortality ratio: This also takes into account the population structure and should be a good indicator. [3]

[6 Marks]

Solution 8.

$$\begin{aligned}
 (aq)_{50}^b &= \int_0^1 {}_t p_{50}^a {}_t p_{50}^b m_{50+t}^h dt \\
 {}_t p_{50}^a &= \exp\left(-\int_0^t m_{50+s}^h ds\right) = \exp\left(-\int_0^t \frac{1}{50-s} ds\right) \\
 \Rightarrow {}_t p_{50}^a &= \exp\left(\left[\ln(50-s)\right]_0^t\right) = \frac{50-t}{50} \\
 {}_t p_{50}^b &= \exp(-0.02t) ; m_{50+t}^h = 0.02 \\
 \therefore (aq)_{50}^b &= 0.02 \int_0^1 \left(1 - \frac{t}{50}\right) \exp(-0.02t) dt \\
 \therefore (aq)_{50}^b &= 0.02 \int_0^1 \exp(-0.02t) dt - \frac{0.02}{50} \int_0^1 t \exp(-0.02t) dt \\
 \int_0^1 \exp(-0.02t) dt &= \left[\frac{\exp(-0.02t)}{-0.02} \right]_0^1 = \frac{1}{0.02} [1 - \exp(-0.02)] = 0.99007 \\
 \int_0^1 t \exp(-0.02t) dt &= \left[t \int \exp(-0.02t) dt \right]_0^1 - \int_0^1 \left(\int \exp(-0.02t) dt \right) dt \\
 \therefore \int_0^1 \exp(-0.02t) dt &= \frac{\exp(-0.02)}{-0.02} - \int_0^1 \frac{\exp(-0.02t)}{-0.02} dt \\
 \therefore \int_0^1 \exp(-0.02t) dt &= -49.00993 + \frac{1}{0.02} \cdot 0.99007 = 0.49357 \\
 \therefore (aq)_{50}^b &= 0.02 \cdot 0.99007 - \frac{0.02}{50} \cdot 0.49357 = 0.0196
 \end{aligned}$$

[6 Marks]

$$\begin{aligned}
{}_{0.5}q_{54.5} &= (1-1/2) q_{55} / (1-1/2 q_{55}) \\
&= 0.5 * 0.003976 / (1-0.5 * 0.003976) \\
&= 0.00199 \\
{}_{5.5}p_{54.5} &= 0.97169 * (1 - 0.00199) \\
&= 0.969756
\end{aligned}$$

(b) Assume that force of mortality is constant across years of age 54 to 55.

$$\begin{aligned}
{}_{0.5}p_{54.5} &= e^{-0.5\mu_{54}} \\
\mu_{54} &= -\ln(1-q_{54}) = -\ln(1-0.003976) = 0.003984 \\
{}_{0.5}p_{54.5} &= e^{-0.5 * 0.003984} = 0.99801 \\
\text{Thus, } {}_{5.5}p_{54.5} &= 0.97169 * 0.99801 = 0.969756
\end{aligned}$$

[6 Marks]

Solution 12.

Method 1:

Let P be the monthly premium and let L be the present value of the loss.

Let X be the random variable representing the number of months for which the policyholder is sick at the end of the month during the policy term of 12 months.

Note that since the transition probabilities are all equal to 0.5, it must therefore follow that $X \sim \text{Binomial}(12, 0.5)$

Now, the total benefit paid out during the policy term is $X(10,000+P)$

The total expenses incurred during the policy term is $0.1P+1,000+12(100+0.01P)$

The total premium received during the policy term is $12P$

Note that since the interest rate is 0% per annum, the present value of the loss is equal to the absolute loss incurred during the policy term.

$$\text{i.e. } L = X(10,000+P) + 0.1P+1,000+12(100+0.01P) - 12P$$

$$\Pr(L > 0) = \Pr\left(X > \frac{11.78P - 2,200}{10,000 + P}\right)$$

$$\Pr(L > 0) < 0.25 \Rightarrow \Pr\left(X \leq \frac{11.78P - 2,200}{10,000 + P}\right) \geq 0.75$$

Since $\Pr(X \leq 6) = 0.61279$ and $\Pr(X \leq 7) = 0.80615$, the least value of P to ensure that the probability of positive loss does not exceed 25% is given by:

$$\frac{11.78P - 2,200}{10,000 + P} = 7$$

$$\therefore P = \frac{70,000 + 2,200}{11.78 - 7} = 15,104.60$$

(OR)

Method 2:

$E(X)=6$ and $V(X) = 3$. Hence, $(X-6)/\sqrt{3}$ is approximately distributed as $N(0,1)$

$$\Pr(X \leq \frac{11.78P - 2,200}{10,000 + P}) \geq 0.75 \Rightarrow \Pr(\frac{X - 6}{\sqrt{3}} \leq \frac{5.78P - 62,200}{1.73205(10,000 + P)}) \geq 0.75$$

Therefore the least value of P to ensure that the probability of positive loss does not exceed 25% is given by:

$$\frac{5.78P - 62,200}{1.73205(10,000 + P)} = \Phi^{-1}(0.75) = 0.67449$$

$$\Rightarrow 5.78P - 62,200 = 1.16825(10,000 + P)$$

$$\Rightarrow P = \frac{11,682.5 + 62,200}{5.78 - 1.16825} = 16,020.49$$

[11 Marks]**Solution 13.**

i) The calculations are shown in the following tables:

Year	Premium	Interest	Mortality Rate	Death Benefit	Expected Death Cost	Independent Surrender Rates	Dependent Surrender rates	Survival Probability
1	500	40	0.00172	5,000	8.60	0.05	0.049914	0.948366
2	0	0	0.00186	5,000	9.30	0.05	0.049907	0.948233
3	0	0	0.00201	5,000	10.05	0	0	0.997990

Year	Surrender Benefit	Expected Surrender Cost	Maturity Benefit	Expected Maturity Cost
1	575	28.70055	0	0
2	650	32.43955	0	0
3	0	0	750	748.4925

Year	Net cash flow	Survival Probability	Profit Signature (ignoring reserves)	Discounted Profit (ignoring reserves)
1	502.70	1	502.70	465.46
2	(41.74)	0.948366	(39.58)	(33.94)
3	(758.54)	0.899272	(682.14)	(541.50)
Total				(109.98)

Hence, the expected present value of loss ignoring reserves is Rs. 109.98. Note that since the reserves earn the same interest rate as that used for discounting the profits, the expected present value of loss is not sensitive to the reserves. Hence, the expected present value of loss allowing for reserves is also Rs. 109.98 [12]

Method 2:

Year	Valuation Mortality Rate	Expected Death Cost - Valuation	End of year reserve after death	End of year reserve before death	Start of year reserves
1	0.00258	12.9	703.25		
2	0.00279	13.95	726.49	738.41	703.25
3	0.003015	15.075	750	762.81	726.49

Year	Reserve at start of year	Interest on Reserve	Reserve required at end of year	Profit vector	Profit Signature	Discounted Profit
1	0	0	666.94	(164.24)	(164.24)	(152.07)
2	703.25	56.26	688.88	28.89	27.40	23.49
3	726.49	58.12	0	26.07	23.44	18.61
Total						(109.98)

ii) a) The required rate of return is about 15% per annum while the expected return is 8% per annum. The absolute loss due to the gap between required and expected return increases with policy tenure.

Thus, the present value of loss is lower for a policy that surrenders in year 1 as compared to a policy that continues till maturity. Hence, the expected present value of profit will increase if surrenders in first year increase. [2]

b) The expected present value of profit will remain unchanged despite the change in valuation interest rate.

Although the emergence of profit will be accelerated as a result of reducing the reserves, as the reserves earn interest at the same rate as that used for discounting the profits, the expected present value of profits is not sensitive to the valuation basis. [2]

[16 Marks]

Solution 14.

i) Let P denote the monthly premium.

EPV [Premiums] =

$$\begin{aligned}
 12 P \ddot{a}_{[30]:35}^{(12)} &= 12 \left(\ddot{a}_{[30]:35} - \frac{11}{24} (1 - v^{35} {}_{35}p_{[30]}) \right) \\
 &= 12 P (15.152 - 11/24 * (1 - 0.1301 * 0.8889)) \\
 &= 12 P * 14.74667 = 176.96 P
 \end{aligned}$$

EPV [Benefits & Expenses] =

$$\begin{aligned}
 &12,500 + 0.5 * 12 P + 0.03 P * 12 \ddot{a}_{[30]:35}^{(12)} - 0.03P + (4,80,000 + 15,000) A_{[30]:35} + 20,000 \\
 &(IA)_{[30]:35}^1 + (7,20,000 - 15,000) v^{35} {}_{35}p_{[30]} \\
 \text{Now, } (IA)_{[30]:35}^1 &= (IA)_{[30]} - v^{35} {}_{35}p_{[30]} ((IA)_{65} + 35 * A_{65}) \\
 &= 2.84718 - 0.1301 * 0.8889 * (5.50985 + 35 * 0.40177) \\
 &= 0.58378
 \end{aligned}$$

$$\ddot{a}_{[30]:35}^{(12)} = 14.74667 \quad \text{as calculated for EPV (Premium)}$$

$$A_{[30]:35} = 0.14234$$

Thus EPV [Benefits & Expenses] =

$$12,500 + 6P + 5.3088 P - 0.03 P + 4,95,000 * 0.14234 + 20,000 * 0.58378 + 7,05,000 * 0.1301 * 0.8889$$

$$= 11.2788 P + 1,76,164.25$$

Equating EPV [Premiums] = EPV [Benefits & Expenses]

$$176.96 P = 11.2788 P + 1,76,164.25$$

$$\Rightarrow P = 1,063.27$$

[7]

ii)

$${}_{30}V^{\text{retrospective}} = (1+i)^{30} / {}_{30}p_{[30]} * (0.97 * 12 P \ddot{a}_{[30]:30}^{(12)} - 4,95,000 A_{[30]:30}^1 - 20,000 (IA)_{[30]:30}^1 + 0.03P - 12,500 - 6P)$$

$$\begin{aligned}(IA)_{[30]:30}^1 &= (IA)_{[30]} - v^{30} {}_{30}p_{[30]} ((IA)_{60} + 30 A_{60}) \\ &= 2.84718 - 0.17411 * 0.935857 * (5.46572 + 30 * 0.32692) \\ &= 0.358514\end{aligned}$$

$$\begin{aligned}A_{[30]:30}^1 &= A_{[30]:30} - v^{30} {}_{30}p_{[30]} \\ &= 0.18283 - 0.17411 * 0.935857 \\ &= 0.019888\end{aligned}$$

$$\begin{aligned}\ddot{a}_{[30]:30}^{(12)} &= \ddot{a}_{[30]:30} - 11/24 (1 - v^{30} {}_{30}p_{[30]}) \\ &= 14.437 - 11/24 * (1 - 0.17411 * 0.935857) \\ &= 14.0533\end{aligned}$$

$$\begin{aligned}\Rightarrow {}_{30}V^{\text{retrospective}} &= 6.1371 * (11.64 P * 14.0533 - 4,95,000 * 0.019888 - 20,000 * 0.358514 + 0.03P - 12,500 - 6P) \\ &= 6.1371 * (156.604 P - 29,514.84)\end{aligned}$$

We know from (i) $P = 1,063.27$

$$\text{Thus, } {}_{30}V^{\text{retrospective}} = 8,47,290.71$$

[7]

[14 Marks]
